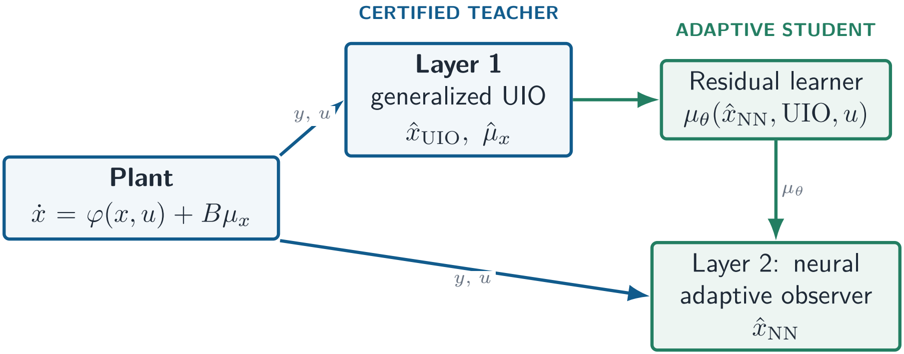
Learning with Unknown Input Observers for Robust Nonlinear Estimation

A hybrid, certified estimator for systems with unknown dynamics

Q. H. Nguyen L. Qi A. Zemouche H. Rafaralahy M. Haddad

IFAC 2026

Global architecture: certified estimation guides learning



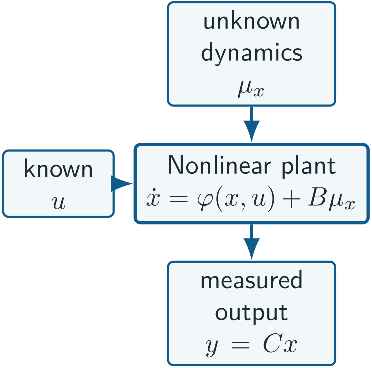
Teacher: bounded, physics-based supervisory signals **Student:** adaptive residual learning

Why estimation becomes difficult with unknown dynamics

Nonlinear plant with an unknown input

$$\dot{x} = \varphi(x,u) + B\mu_x, \qquad x \in \mathbb{R}^{n_x},$$
$$y = Cx, \qquad \mu_x \in \mathbb{R}^{n_\mu}.$$

- $\varphi(x,u)$: known nonlinear physics;
- μ_x: unmodelled dynamics, disturbances, or force residuals;
- only y is measured and μ_x must be inferred online.



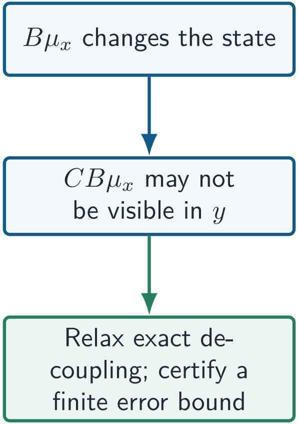
Goal: estimate x and infer μ_x.

The classical UIO condition is often too restrictive

Direct unknown-input decoupling requires

$$\mathrm{rank}(CB) = \mathrm{rank}(B).$$

- This guarantees that the output is informative enough to isolate μ_x.
- In real vehicle and robotic systems, CB can be rank deficient.
- Enforcing exact decoupling can then rule out an otherwise useful observer.



Regularization converts a rank obstacle into a bounded disturbance

$$
\begin{aligned}
\dot{x} + E_{\delta_1}\dot{\mu}_x &= \varphi(x,u) + B\mu_x + E\omega_{\delta_1}, \\
y &= Cx + D_{\delta_2}\mu_x + D\omega_{\delta_2}, \\
E_{\delta_1} &= \delta_1 E, \qquad D_{\delta_2} = \delta_2 D, \\
\omega_{\delta_1} &= \delta_1 \dot{\mu}_x, \qquad \omega_{\delta_2} = -\delta_2 \mu_x .
\end{aligned}
$$

- The transformed model is equivalent to the original plant.
- The tunable δ_1, δ_2 restore observer design freedom.
- The price is explicit: ω_δ enters the performance analysis.

Certified consequence

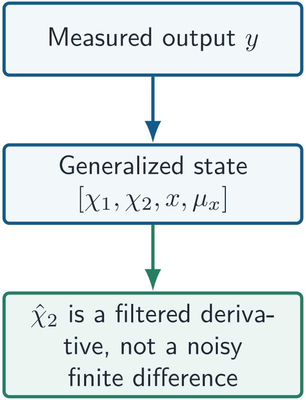
$$
\left\|\tilde{\zeta}\right\|_{\mathcal{H}^1} \leq \lambda_\delta \left\|\mu_x\right\|_{\mathcal{H}^1} + \beta \left\|\xi_0\right\| .
$$

Exact convergence is recovered when the classical rank condition holds.

Output derivatives create a generalized unknown-input observer

Introduce $\chi_1 = y$ and $\chi_2 = \dot{y}$ in the augmented state $\zeta = \begin{bmatrix}\chi_1^\mathsf{T} & \chi_2^\mathsf{T} & x^\mathsf{T} & \mu_x^\mathsf{T}\end{bmatrix}^\mathsf{T}$:

$$\begin{aligned}
\dot{\chi}_1 &= \chi_2, \\
\dot{\chi}_2 &= C\bar{\varphi}_x(x,u) + C\frac{\partial \varphi}{\partial x}(x,u)B\mu_x + CB\dot{\mu}_x, \\
\dot{x} &= \varphi(x,u) + B\mu_x, \\
y_\chi &= \chi_1 + C_\chi \chi_2 + Cx.
\end{aligned}$$



Layer 1: LMI-certified generalized UIO

Observer realization

$$\dot{\eta} = P_\zeta \varphi_\zeta(\hat{\zeta}, u) + K(y - \mathcal{H}\hat{\zeta}),$$
$$\hat{\zeta} = \eta + Q_\zeta y, \qquad P_\zeta \mathcal{E} + Q_\zeta \mathcal{H} = I.$$

$$K = \mathcal{P}^{-1}\mathcal{Q}^{\mathsf{T}}.$$

Guarantee

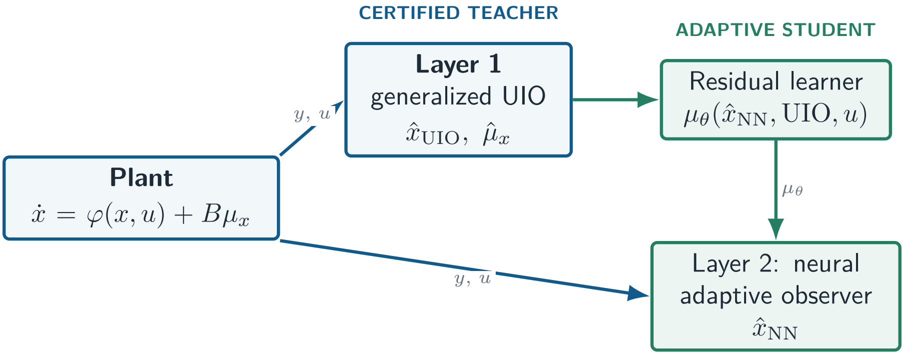
Feasibility yields an $\mathcal{H}^1$ estimation bound for the state and unknown input.

For every vertex j of the nonlinear embedding, solve

$$\begin{bmatrix} \Xi_j & \Gamma_j & \Theta_j \\ \Gamma_j^{\mathsf{T}} & -\lambda I_{2n_\mu} & \Gamma_j^{\mathsf{T}} \\ \Theta_j^{\mathsf{T}} & \Gamma_j & -\epsilon^{-1}\mathcal{P} \end{bmatrix} \prec 0.$$

- offline convex synthesis;
- online observer with a quantified robustness level λ_δ.

A certified teacher supervises learning



Teacher: bounded, physics-based supervisory signals **Student:** adaptive residual learning

Layer 2: a neural observer estimates its own state

State-dependent residual model

$$\mu_\theta(\hat{x}_{\mathrm{NN}}) = \mathcal{N}_\theta(\hat{x}_{\mathrm{NN}}, \hat{x}_{\mathrm{UIO}}, \hat{\mu}_x, u)\,.$$

Layer-2 observer dynamics

$$\begin{aligned}\dot{\hat{x}}_{\mathrm{NN}} &= \varphi(\hat{x}_{\mathrm{NN}}, u) + B\mu_\theta(\hat{x}_{\mathrm{NN}}) + L_{\mathrm{NN}}(y - y_{\mathrm{NN}}),\\ y_{\mathrm{NN}} &= C\hat{x}_{\mathrm{NN}}.\end{aligned}$$

The UIO supplies supervisory inputs; it does not replace the independent dynamic state $\hat{x}_{\mathrm{NN}} \in \mathbb{R}^{n_x}$.

Three contributions to $\dot{\hat{x}}_{\mathrm{NN}}$

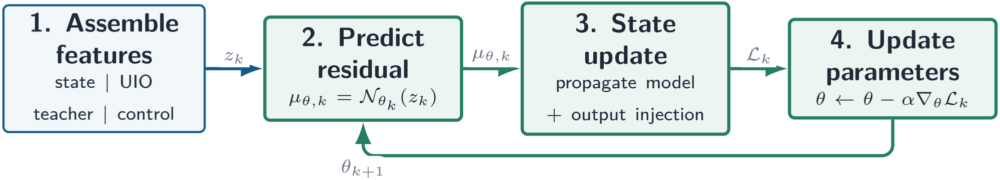
- known nonlinear physics φ;
- learned unknown dynamics $B\mu_\theta$;
- measurement innovation $L_{\mathrm{NN}}(y - y_{\mathrm{NN}})$.

Certified error channel

With $\epsilon_{\mathrm{NN}} = x - \hat{x}_{\mathrm{NN}}$ and $\Delta_x = \mu_x - \mu_\theta$,

$$\begin{aligned}\dot{\epsilon}_{\mathrm{NN}} &= \left(\nabla_x^\varphi - L_{\mathrm{NN}}C\right)\epsilon_{\mathrm{NN}} + B\Delta_x\,,\\ \|\epsilon_{\mathrm{NN}}\|_{\mathcal{L}_2} &\le \lambda_{\mathrm{NN}}\,\|\Delta_x\|_{\mathcal{L}_2} + \beta_{\mathrm{NN}}\,\|\epsilon_{\mathrm{NN}}(0)\|\,.\end{aligned}$$

Layer 2: online parameter adaptation



Composite objective at sample k

$$
\begin{aligned}
\mathcal{L}_k = {} & \|\hat{x}_{\mathrm{NN},k} - x_{\mathrm{ref},k}\|^2_{T_{\mathrm{ref}}} + \|y_k - C\hat{x}_{\mathrm{NN},k}\|^2_{T_y} \\
& + \|\hat{x}_{\mathrm{NN},k} - \hat{x}_{\mathrm{UIO},k}\|^2_{T_{\mathrm{UIO}}} + \lambda \|\mu_{\theta,k} - \hat{\mu}_{x,k}\|^2 .
\end{aligned}
$$

The four terms respectively promote reference tracking, measurement consistency, agreement with the UIO, and residual matching to the teacher proxy.

Features and parameter update

$$
\begin{aligned}
z_k &= \mathrm{col}(\hat{x}_{\mathrm{NN},k}, \hat{x}_{\mathrm{UIO},k}, \hat{\mu}_{x,k}, u_k), \\
\mu_{\theta,k} &= \mathcal{N}_{\theta_k}(z_k), \\
\theta_{k+1} &= \theta_k - \alpha_k \nabla_\theta \mathcal{L}_k, \qquad \alpha_k > 0.
\end{aligned}
$$

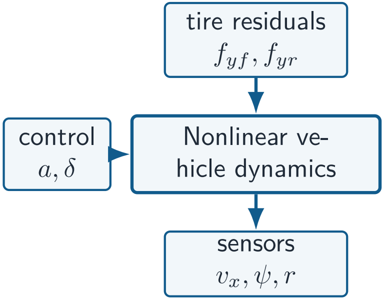
The LMI-designed observer gain L_{NN} remains fixed while the network parameters adapt online to reduce Δ_x.

Case study: nonlinear vehicle with unknown tire forces

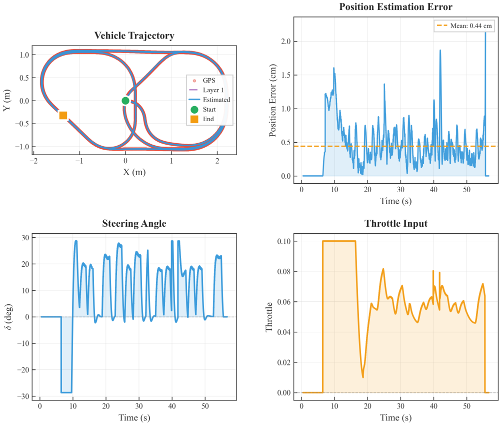
Vehicle state, input, and measurement

$$
\begin{aligned}
x &= \begin{bmatrix} v_x & v_y & \psi & r \end{bmatrix}^{\mathsf{T}}, \qquad & u &= \begin{bmatrix} a & \delta \end{bmatrix}^{\mathsf{T}}, \\
y &= \begin{bmatrix} v_x & \psi & r \end{bmatrix}^{\mathsf{T}}, \qquad & \mu_x &= \begin{bmatrix} f_{yf} & f_{yr} \end{bmatrix}^{\mathsf{T}}.
\end{aligned}
$$

- Lateral velocity v_y is not directly measured.
- Front/rear lateral tire-force residuals represent the unknown input.
- A layered controller provides the manoeuvring input used by both estimators.



Trajectory tracking: centimetre-scale position error



Observed outcome

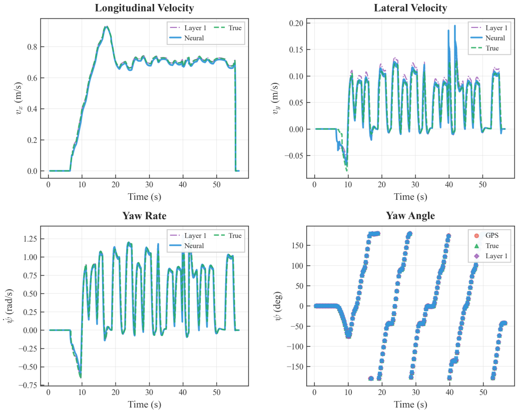
The estimated trajectory closely follows the GPS reference during the manoeuvre.

Reported metric

Mean position error: **0.44 cm**.

The lower panel confirms feasible steering and acceleration commands.

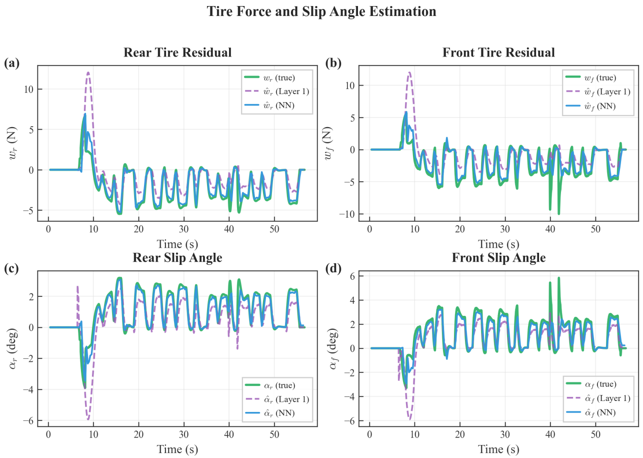
State estimation: the unmeasured lateral velocity is reconstructed



What matters

- Measured states remain consistent with their sensor channels.
- The estimator supplies v_y, which is absent from y.
- The learned observer refines transient performance beyond its certified teacher.

Residual learning improves tire-force reconstruction



Interpretation

- Layer 1 gives a bounded, physically meaningful residual estimate.
- Layer 2 converges faster and tracks tire-force residuals more closely.
- The learned map captures state-dependent residual dynamics.

Key takeaways

1. **Relaxed UIO design:** regularization and output derivatives remove reliance on exact unknown-input decoupling.
2. **Certified robustness:** LMIs provide an explicit $\mathcal{H}^1$ bound for the generalized observer and an $\mathcal{L}_2$ bound for the neural observer.
3. **Reliable learning:** the UIO supplies stable supervisory signals; the neural observer learns the residual dynamics online.
4. **Vehicle evidence:** accurate trajectory tracking, recovery of the unmeasured lateral state, and improved tire-force estimates.

Central message

Certification and learning are complementary: the observer makes adaptation trustworthy, and adaptation makes the observer more accurate.